

A Cross-Cultural Study of Relational Maintenance in Tech Caregiving

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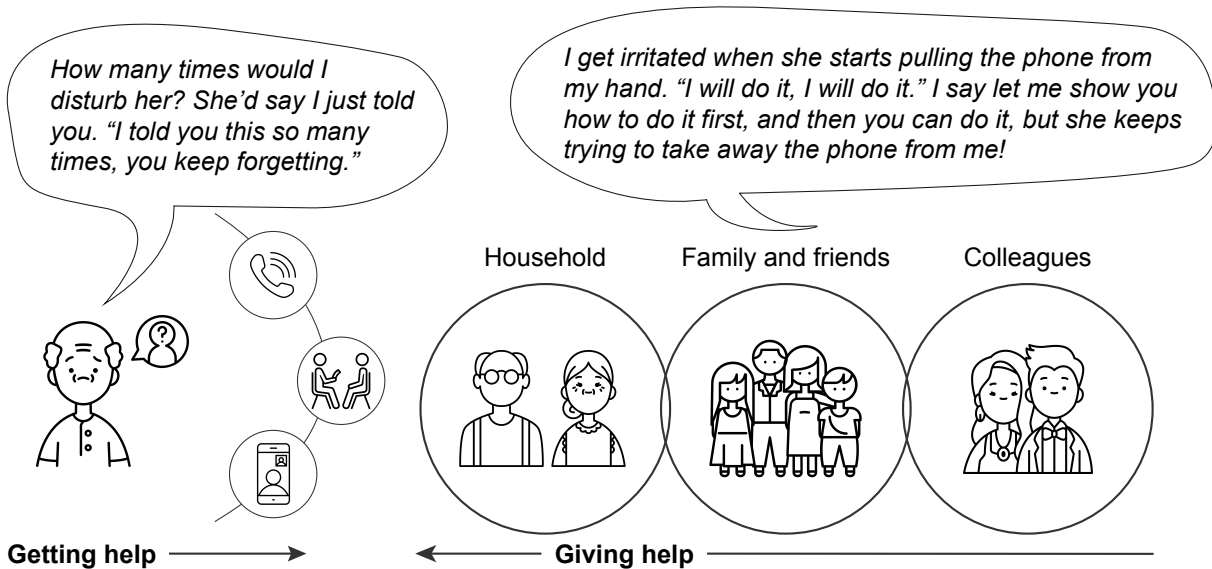


Figure 1: Older adults prefer getting tech support from family, friends, and colleagues in day-to-day life—whether in person or remotely. Using the *relational maintenance strategies framework* [9], we examine how tech caregiving challenges social relationships and what maintenance strategies are employed to mitigate them.

ABSTRACT

In day-to-day life, older adults prefer getting tech support from loved ones, rather than relying on instruction manuals or institutional support. But asking for help, frequently or repeatedly for similar issues, can strain relations between older adults and their helpers. This paper examines how relationships are maintained when younger helpers give and older adults get tech support—across three different cultures: North American, South Asian, and Middle Eastern. Our deductive application of the *relational maintenance strategies framework* to 40 in-depth, semi-structured, and cross-cultural interviews identify challenges in tech caregiving, like avoiding the topic or interaction, and mitigation efforts, like directly or indirectly assuring each other of their skills and abilities. We discuss how social interactions are managed around tech support, how individual constraints, like time, place, or abilities are respected,

and how self-conscious emotions, like guilt, embarrassment, or empathy are handled.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Empirical studies in accessibility**.

KEYWORDS

older adults, tech caregiving, tech support, accessibility, relational maintenance

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1 INTRODUCTION

Technology (tech) support has become a common type of informal caregiving for people with relatives, friends, or colleagues who are less comfortable with personal technology use, such as computers, tablets, or smartphones [36, 41]. Who hasn't spent hours

on the phone with a loved one trying to solve issues on an unseen device, or repeating instructions for the same task for the umpteenth time sitting beside them? Particularly for older adults in their lives. Older adults use conventional tech applications (apps) much more than any *senior-friendly* variants that are designed specifically for them [22, 23]. But as apps keep having more features and more sophisticated navigation patterns, older adults keep encountering more and more unique challenges in learning and using them [26, 27, 43, 51, 52]. And so increases the need for go-to tech helpers. Helpers, who are in their social network, are much more preferred by older adults than instruction manuals or tech helplines [24, 36]. Even when caregivers want to help out, this "tech caregiving" relationship has its challenges. For instance, helpers may be unavailable or remotely located ("When my dad has an issue with a device now, we have to wait until I or one of my siblings travel home, or we have to find a neighbor who can help.", [41]) while older adults may avoid or postpone seeking help because of a perceived negative attitude or feelings of guilt for troubling others ("I asked a friend to show me, and he actually had to show me twice because he showed me and then I forgot, and I couldn't do it again.", [36]).

Older adults' preference toward getting tech support socially, is new [36]. Early studies examining older adults' technology adoption and technology learning reported that facilitating conditions, such as instructional manuals and computer support, determine older adults' intention to use a technology [2, 42], and that most older adults prefer structured training materials despite difficulties using them [25]. Recent studies, however, report a preference toward step-by-step instructions provided by another trusted person and trial-and-error, while relying less on instruction manuals [2, 36, 46]. Although evidence toward a preference for trial-and-error methods when onboarding, learning, or troubleshooting remains mixed [21, 30, 36, 51], seeking tech support socially has become commonplace [36, 41]. Thus, tools facilitating this type of tech support for older adults are currently being explored [1, 18, 49, 53]. Older users' reservations in turning to their social network for tech support, and tech helpers' frustration to provide such support efficiently and effectively have been briefly discussed elsewhere [36, 41]. But a systematic examination of this tech caregiving relationship—which can greatly influence the design of tech caregiving tools—remains under-explored.

Much of this tech support can happen around mobile devices, such as smartphones, tablets, and wearables. American Association of Retired Persons (AARP), the leading advocacy group for older adults in the U.S., reported a significant rise in the use of smartphones and tablets in 2020 for activities, such as making online purchases, ordering groceries, banking, and engaging health services, which continued in 2021 [20]. In spite of that, two in five 50-plus adults reported feeling technologies are not designed with them in mind, citing the apps' complexities [20]. These usability issues may also vary across cultures. In a recent survey, the country of residence explained 14% of variance in predicting general technology use during the COVID-19 pandemic [14].

On the other hand, more and more services essential to daily life are being offered via sophisticated mobile apps. Thus, a grim future—where older adults move into assisted living because they are unable to use online-only government services—may not be far away [22]. So, it is not surprising that more older adults now

own mobile devices than ever before—in 2021, 61% of 65-plus and 83% of 50-plus U.S. adults owned a smartphone [38]. Consequently, older adults are seeking more tech support for mobile devices from their social network [25, 36]—which is the context within which we situate our research questions:

*How does tech caregiving challenge social relationships?
What maintenance efforts and strategies are employed
to mitigate those challenges? And how do these behaviors
differ across different cultures?*

Caregiving is a complex social practice where one takes care of their loved one. Caregivers may perform a wide variety of tasks to help care recipients in their daily lives [3]. Caregiving can occur at different stages of life and for different reasons, such as an onset of illness, like Dementia, or during periods of physical or mental disabilities [8, 16, 40]. Much prior work in this area focuses on the role of caregivers in supporting different activities [10, 28], caregiver burden [8, 15, 31, 47], and the effects of new technologies on caregiving work [5, 10, 13, 29, 33]. A few HCI studies have looked into how caregivers and care recipients interact in the context of digital technologies and identified that work as *technology caregiving* [39]. Tech caregiving has been studied in supporting online activities for adults with cognitive impairments [39], safeguarding against cybersecurity and privacy threats for older adults [34] and people with mild cognitive impairments (MCI) [32], and using augmentative and alternative communication (AAC) systems by people with amyotrophic lateral sclerosis (ALS) or cerebral palsy (CP) [12].

Like all other types of caregiving, tech caregiving to older adults can inflict stress and impact the caregiver–care recipient relationship. So when do these breakdowns happen, and when they do, how are they mitigated? In a qualitative study, we explore these questions across three different cultures (North American, South Asian, and Middle Eastern) through a *relational lens* and identify exemplars of positive and negative conflict mitigation strategies. Our analysis uses the *relational maintenance* strategies framework that identifies ten approaches to maintain relationships, such as avoidance, assurances, and the use of mediated communication [9]. Future tech caregiving tools for older adults can use our findings to prevent negative and facilitate positive relational maintenance behaviors.

2 METHODS

We conducted a set of semi-structured interviews and online surveys with older adults (ages 60 and over) and younger helpers. The two groups were recruited and interviewed separately, but some of them were related as family members ($n = 9$ dyads). Our older participants owned and used a current mobile device (e.g., smartphone, tablet, or smartwatch) at least once a week. Younger participants regularly assisted one or more older mobile users at least once a week. Interviews were conducted separately for younger helpers and older adults, and remotely via phone or video-calling apps (e.g., WhatsApp or Skype) with permanent residents of three different countries, United States, Iran, and India. The study was approved by a university-wide institutional review board and recruitment occurred via flyers on social media, emails, and word of mouth.

Interviews were audio recorded and focused on how older adults learn to use mobile technologies and seek help when problems crop

up. We asked about their preferences and methods when asking for tech support socially—like if they have somebody who helps them learn or solve tech problems, how these people help them out, and how they perceive these interactions and challenges they encounter in seeking help socially. With younger participants, our interview focused on how they supported older users in learning and troubleshooting mobile technologies, what materials they used in different social settings, how they participated in those social interactions, what challenges they faced, and what strategies they used when older users needed help frequently. The interview script is provided in the Appendix. In addition, we recorded the demographics of older and younger participants and the mobile device proficiency of older adults. Proficiency was measured using the 14-item Mobile Device Proficiency Questionnaire (MDPQ) [37], which is a self-reported measure of smartphone skills and has been validated with older adults (Appendix, Table 4). MDPQ scores range from 14 (beginner) to 70 (proficient).

Study sessions lasted about an hour and were conducted in 2021. Interviews were conducted in participants' preferred language, English, Farsi, or Bengali. The audio corpus was 20 hours and 28 minutes long, and deidentified for storage and analysis. The first author (a native Farsi speaker) transcribed and translated the interviews from the U.S. and Iran and the second author (a native Bengali speaker) transcribed and translated the interviews from India. Transcribed interviews were then imported to *ATLAS/ti* for thematic analysis [6, 7]. We conducted a deductive analysis using a comprehensive typology of 10 relational maintenance strategies [9]: positivity, openness, assurances, social network, shared tasks, joint activities, use of mediated communication, avoidance, anti-social, and humor. We adapted the strategy definitions to tech support and included modern communication methods, such as online activities. Table 5 (Appendix) details our codebook.

3 RESULTS

3.1 Participants

Twenty three older adults (11 females, $Mdn_{age} = 67$, $Mdn_{MDPQ} = 39$) and 17 younger helpers (9 females, $Mdn_{age} = 26$) participated in our study (Table 1). As per MDPQ-14 scores, our older participants were neither absolute beginners nor experts. They reported using either smartphones or tablets or both. Nineteen older adults were college educated, one had a 2-year degree, three had a high school diploma, one had less than a high school diploma, and one had a master's degree.

3.2 Tech caregiving challenges

Our deductive analysis identified both *avoidance* and *anti-social* strategies in managing social interactions around tech support. However, the same strategies manifested in different types of behaviors in older adults and tech helpers. Older adults may *avoid the topic* of tech altogether, because of disinterest or privacy, *postpone* the interaction (around tech support) to a time more amenable to the tech helper, or *limit* their help-seeking based on interaction history with a tech helper or self-conscious emotions like guilt or embarrassment. Younger helpers, on the other hand, may *avoid the interaction* but solve the issue, or *postpone* it for another time because they are unavailable, they don't know how to support, or the

issue needs face-to-face interactions and they are communicating via phone or text. Sometimes interactions were not avoided but instead, they became *directly* or *indirectly* unfriendly or forced. These behaviors, their percentages, and example quotes are presented in Table 2.

3.3 Relational maintenance in tech caregiving

Despite how both older adults and tech helpers engaged in avoidance behaviors (Table 2), different strategies were used to maintaining their social relationships during tech caregiving. We identified five major strategies that can be grouped into two broader categories to contextualize the cross-cutting behaviors, further emphasizing how interactions around tech caregiving were changed, improved, paced, or planned.

3.3.1 Social network, joint activities, and the use of mediated communication. We found that older adults relied heavily on their social network for tech support—namely, family (in the same household or distributed), friends, and colleagues, and used mediated communication tools, such as calls, video calls, and texts frequently, in addition to in-person interactions (Figure 1). To ensure they are not bothering one person multiple times or postponing the problem until they are available, older adults had a small reserve of people they could reach out for help.

"I have a brother and a nephew who are good with phones, and my daughter is very experienced. I learn by asking questions of these people. If they are not available I ask a friend of mine." –IR-P4, 61M

"Of course, in general, I ask the kids. They are much more knowledgeable about mobiles. They know these things. I'd ask my nephew, my son. My younger son knows a lot of things, I'd ask him." –IN-P6, 64M

On the other hand, if helpers did not know how to help, they would also identify another person within the older adult's social network to solve the issue.

"My parents wanted to project my video on the TV ... My brother video-called me and showed me what they were doing and I guided through the process." –IR-P4, 24M

Although the focus of our study was younger tech helpers, we found that the tech helpers in older adults' social networks were not *exclusively* younger individuals. Sometimes, older adults provided support to each other.

"I help out a lot of people ... a lot of people my age ... my friends and acquaintances. Like go here, do this, do that." –IN-P4, 69M

"Whatever help I need, it is my brother. If anything happens, I reach out to my brother [70M]. I talk to him ... he tries to make me understand. Often I can do things after learning them from him. And ... sometimes I can't. Then when I am near him [in person] he would teach me those." –IN-P3, 73F

Tech caregiving occurred via mediated communication frequently, in spite of the associated challenges like not being able to see the

Table 1: Participant demographic per country ($n = 40$).

Country	Cohort	Males	Females	Age	MDPQ-14
India	Older adults	4	5	<i>Mdn</i> = 64 yrs., <i>SE</i> = 1.57	<i>Mdn</i> = 33, <i>SE</i> = 4.55
	Tech helpers	1	4	<i>Mdn</i> = 28 yrs., <i>SE</i> = 4.33	-
Iran	Older adults	3	4	<i>Mdn</i> = 67 yrs., <i>SE</i> = 2.86	<i>Mdn</i> = 39, <i>SE</i> = 5.48
	Tech helpers	3	2	<i>Mdn</i> = 26 yrs., <i>SE</i> = 5.96	-
U.S.	Older adults	5	2	<i>Mdn</i> = 67 yrs., <i>SE</i> = 3.29	<i>Mdn</i> = 54, <i>SE</i> = 5.23
	Tech helpers	4	3	<i>Mdn</i> = 26 yrs., <i>SE</i> = 1.94	-

care receiver's device screen or trying to understand the issue without the complete context.

"Some things he might have been doing at home and getting stuck, and then he'd call me, call me from another phone and tell me that this is where I am getting stuck and what should I do. I've often told him over the call, do this, this step after that step." –IN-P7, 60M

When older adults and helpers resided in the same household (or visited often), tech caregiving happened organically, when spending time together (joint activities):

"My wife and I sit next to each other when using the phone and tablet, and if I get stuck on something I ask her. She is around more often than my kids." –US-P7, 71M

"I don't need to teach my mother much. I spend most of my time with my mother, so she watches me using the phone all the time." –IN-P5, 28F

3.3.2 Assurance and openness. Both giving and getting support can produce negative emotions like frustration, impatience, and annoyance among helpers and disappointment, anxiety, guilt, and embarrassment among older adults, respectively (Table 2). Results revealed that helpers mitigate those challenges by practicing *empathy*, having *patience*, and actively *listening* to older adults' problems. Furthermore, older adults and helpers assured each other of their abilities and skills, respectively, directly or indirectly. These behaviors and example quotes are presented in Table 3.

3.4 Cross-cultural observations

When examining tech caregiving challenges, maintenance strategies, and behaviors across the three cultures, we identified more similarities than differences. Challenges, manifested as *avoidance* and *anti-social* behaviors, were reported across cultures. But these behaviors were more commonly reported by Iranian (50%) and Indian (71%) than U.S. participants (35%), which may have stemmed from the differences in older adults' living arrangements in the three respective countries. 100% of our U.S. participants reported living independently, which is common in developed countries across North America and Europe [35]. In contrast, 100% of our Indian participants reported living with family, and thus, spending more time with family members who might take on the role of tech caregivers. One helper put it this way:

"I don't know if anyone puts in so much effort to help out their parents. Because I've never heard anyone explain these things to their parents. I rarely go out with my

colleagues and spend most of my free time with my parents." –IN-P5, 28F

Although all our Iranian older adults reported living independently [45], they discussed spending time with family and friends (86%) more than Indian (44%) and U.S. participants (42%). Similarly, only 3/7 U.S. older adults preferred asking for help socially compared with 7/9 Indian and 7/7 Iranian older adults.

Older adults and helpers from all three countries reported using mediated communication strategies, like phone calls, video calls, and texts. However, Indian (71%) and Iranian (50%) participants discussed remote communication around tech caregiving more than U.S. participants (42%). Unique to the Indian participants was the use of a flip phone when seeking support from family or friends remotely. 5/14 Indian participants reported using a flip phone when giving or getting help with a smartphone issue. Older adults reported using a flip phone to get support even when they had access to other smartphones in their household. In comparison, Iranian participants reported using another smartphone when trying to help remotely.

"I bought my parents an older model, a flip phone. I've tried to teach her how to use smartphone apps while I am on the flip phone giving instructions, e.g., when I initiate a video call using the Jio App." –IN-P3, 28M

4 DISCUSSION

Tech caregiving is becoming an essential type of caregiving globally—as the aging population rises sharply, more services offered digitally, and families spread out geographically. Although HCI researchers began exploring the challenges that older adults encounter with computer technologies as early as circa 2000 [11, 17, 44], tech caregiving to older adults has drawn research attention only recently [32, 34, 39]. Especially, since older adults are rarely adopting simplified, senior-purpose apps [23, 48, 50], and would rather use technologies with family and friends across generations. In contrast to findings from earlier studies, older adults now prefer to troubleshoot technologies with people in their social network, not instruction manuals [19, 36].

Like general caregiving can cause caregiver burden or burnout [8] and impact social relationships [12], tech caregiving is expected to add some stress to existing relationships. Motivated by that premise, we examined how tech caregiving challenges relationships, and what the common mitigation strategies are. Challenges manifested as avoidance and anti-social behaviors—more among older adults and helpers who spent more time together (Table 2). To mitigate those challenges, multiple tech helpers were recruited by older

Table 2: Tech caregiving challenges, example behaviors, and percentage of sample reporting (IN-, IR-, and US- refer to Indian, Iranian, and U.S. participants, respectively. M and F refer to male and female participants, respectively.)

Cohort	Example behaviors	Example quotes
Avoidance: Evasion of interaction or issues (47.5%)		
Older adults	Avoid topic	"When I got my father a phone, I wanted to teach him everything, like I use a smartphone. But my father was like: 'Do you want me to forget the little I've learned?'" –IN-P5, 28F "Yes, I can ask other people ... ask in one of my WhatsApp groups how they have solved the problem. But then I'm making my troubles public which I don't wish to." –IN-P2, 70M
	Postpone support	"She saves a lot of her queries and things she wants to learn over the week for the weekend. When Saturday or Sunday arrives, she'll go 'Are you free now?'" –IN-P5, 28F "I call him [my brother] up according to his schedule ... when he might be available ... I don't contact him right away, because he is not always free. When he has time, I call him." –IN-P3, 73F
	Limit support	"Now how many times would I bug her. She has her studies ... disturbing her all the time is not right." –IN-P9, 69F "I Google or I text one of my kids how to do something. Googling is more effective but I prefer to ask my kids because it makes them communicate with me. They explain it to me in a way I understand, but I only ask once. They won't explain it to me more than once." –US-P6, 66F "I'd think [...] she is such a young girl and she can do everything while I am such a grown adult and I can't do anything ... it's such a shame." –IN-P9, 69F
Helpers	Avoid interaction	"When I have to scan a document and send it to someone via chat, he [son] does that on his own. He hasn't taught me anything." –IN-P8, 61F "I take my mother's tablet, fix it and give it back." –IR-P1, 51F
	Postpone support	"I used to send her to the neighbors when I was away, so that they can fix it for her." –IR-P1, 51F "After trying two to three times, I have to say leave it. When I am there I will show you in person." –IN-P3, 28M
Anti-social: Behaviors which seem unfriendly or coercive (25%)		
Older adults, helpers	Direct	"I get irritated when she starts pulling the phone from my hand ... 'I will do it ... I will do it.' I say let me show you how to do it first ... and then you can do it ... but she keeps trying to take away the phone from me!" –IN-P1, 43F "When I ask her did you learn, sometimes she says 'let it go' or 'whatever.'" –IR-P1, 51F
	Indirect	"I get frustrated when there are a number of intermediate steps in a task, when I repeat things a number of times but they don't get it [...] 'where is it written?', 'what is written?'" –IN-P2, 18F "If she doesn't say anything after I try to teach her, I assume she has learnt or she is too annoyed with it and just wants to leave." –IR-P1, 51F

adults from their social network (as also reported in [21, 36]), problems were brought up organically during joint activities, mediated communication tools (like phone and video calls) were frequently used, assurances were provided about their skills and abilities, and empathy and patience were practiced (Table 3). Mitigation strategies were discussed (85%) more than caregiving challenges (52%),

and there lies a design opportunity for solutions that can reduce the negative relational effects of tech caregiving.

We did not interview older adults and helpers together as a dyad, but separately to make sure they can openly discuss their challenges when getting and giving support, respectively, without hurting each other's feelings. Other types of studies, like a direct observation of tech caregiving, may identify issues and aspects

Table 3: Example behaviors of assurance and openness in tech caregiving, and percentage of sample reporting. (IN-, IR-, and US- refer to Indian, Iranian, and U.S. participants, respectively. M and F refer to male and female participants, respectively.)

Cohort	Example behaviors	Example quotes
Assurance: Covertly and overtly assuring each other of their skills and abilities (12.5%)		
Older adults	Direct	"I learn by asking questions from others: children, siblings. Mashallah, all the kids know everything now. [...] I feel confident to ask others who know more about phones" –IR-P3, 60F
	Indirect	"When I do go out we go as a family and my daughter always has a phone. [...] My daughter checks the weather and uses Waze to help us find the way. She is like a 'talking computer' for us." –IR-P3, 60F
Helpers	Direct	"They forget and I have to repeat it and remind them. I tell them you know how to do it." –US-P0, 21M
	Indirect	"My grandmother feels better when I help her through a video call, because she knows if she does something wrong, then I am there to fix it." –US-P3, 26F
Openness: Direct discussions and listening to one another (17.5%)		
Helpers	Empathy	"I feel that it's really hard for them and also I get it that they are not that happy about coming to me with questions. This is why I don't want to make them feel bad." –US-P3, 26F
	Patience	"The main thing is that you need to be patient. After I show them I ask them to do it by themselves, and ask them to ask questions. I will have to answer the questions patiently." –US-P3, 26F
	Listening	"I show them for a few days, and then I go back with some follow-up questions like did you do that? Do you know how to do this? Is there something else you wanna do and want to ask me how?" –US-P3, 26F "I wish I could do better to make sure they have learned. At the moment when I'm with her, I have her do it, but if I'm not there I try to listen to her and her feedback to see how she did." –US-P5, 31F

that we did not. Nevertheless, dyadic interviews in another study revealed some of the same challenges as we did, such as repeatedly needing instructions from friends or family [36]. We did not find all types of relational maintenance strategies in our study, such as humor or positivity—probably because in our interviews participants recalled past events and actions. Future direct observations of tech caregiving may find these relational maintenance strategies in use.

Although we focused on younger helpers, we found some older adults taking on the role of tech helpers. These older helpers set themselves apart from their less fluent peers: *"I'd never ask older people. They don't know anything about these things. Poor folks, they don't know anything."* –IN-P6, 64M. This is in line with the idea of *situated elderliness*—which advocates that situational aspects of environments, objects, and life changes can affect an older adult's perception of the self as "elderly" or "not" [4]. In conclusion, we identified some pain points of tech caregiving to older adults, and common mitigation behaviors currently in practice. Our findings can benefit the design of future tech caregiving tools, like interactive tutorials with humans-in-the-loop [18].

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A RESEARCH INSTRUMENTS

A.1 Interview script for older adults

- What kind of mobile devices do you use?
- How long have you used it? How often do you use it?
- What are the five applications that you most frequently use on your device?
- Which one(s) are easier for you to use?
- What is an app you use more regularly?
- What is an app that you'd like to learn better?
- What different things do you do with the apps? What are the things you do frequently? What are the things you do rarely?
- Have you ever found it difficult to do (one of the rare things)?
- How did you learn how to use it? Did someone help you? Did you do it yourself? Do you still ask for help to use app X? For how long did you need help to learn it in the first place?
- (if multiple methods were used) Which one do you think was the most effective to learn and to troubleshoot?
- Have you ever tried the help option in app x? If yes, was it useful?
- Do you think you would have succeeded if you had more help? What kind of help would you prefer?

A.2 Interview script for helpers

- How do you usually help your care receivers in using mobile applications? Give us some examples, please.
- Do you help them when you are physically there or when away? How is it different?
- How does the whole process of helping work? What do you do, and what do they do?
- How do you make sure they have learned or solved the issue?
- How frequently do these issues occur and do they need your help?
- What types of tasks are usually difficult for them that they ask for help from you?
- Do they try other things before coming to you for help?
- How often do you help them with the same or similar tasks?
- Could you tell us some apps you tried to teach them, that they rarely/never use? Why do you think that is the case?
- What are some apps you tried teaching them and they are using it?
- What is the most frustrating (annoying) thing during this process?
- How do you tackle those frustrations?

A.3 Mobile Device Proficiency Questionnaire-14 (MDPQ-14)

Table 4: MDPQ-14 was used to assess older adults' mobile device proficiency.

Hello there, thank you for your participation!

The questions below ask about your ability to perform a number of tasks with a mobile device.

Please answer each question by selecting the option that you feel is most appropriate.

If you have never tried to perform a task with a mobile device or do not know what a task is, please choose

“Never tried,” regardless of whether or not you think you may be able to perform the task.

Please remember that you are rating your ability to perform each of these tasks specifically using a mobile device (tablet or smartphone).

1. Using a mobile device I can adjust the screen brightness	1	2	3	4	5	
Never tried	☐	☐	☐	☐	☐	Very easily
2. Using a mobile device I can connect to a WiFi network						
Never tried	☐	☐	☐	☐	☐	Very easily
3. Using a mobile device I can open Email						
Never tried	☐	☐	☐	☐	☐	Very easily
4. Using a mobile device I can view pictures sent by Email						
Never tried	☐	☐	☐	☐	☐	Very easily
5. I can transfer information (files such as music, pictures, documents) on my computer to my mobile device						
Never tried	☐	☐	☐	☐	☐	Very easily
6. Using a mobile device I can store information with a service that lets me view my files from anywhere (e.g., Dropbox, Google Drive, Microsoft Onedrive)						
Never tried	☐	☐	☐	☐	☐	Very easily
7. Using a mobile device I can find information about local community resources on the Internet						
Never tried	☐	☐	☐	☐	☐	Very easily
8. Using a mobile device I can find information about my hobbies and interests on the Internet						
Never tried	☐	☐	☐	☐	☐	Very easily
9. Using a mobile device I can enter events and appointments into a calendar						
Never tried	☐	☐	☐	☐	☐	Very easily
10. Using a mobile device I can set up alerts to remind me of events and appointments						
Never tried	☐	☐	☐	☐	☐	Very easily
11. Using a mobile device I can watch movies and videos						
Never tried	☐	☐	☐	☐	☐	Very easily
12. Using a mobile device I can read a book						
Never tried	☐	☐	☐	☐	☐	Very easily
13. I can update games and other applications on a mobile device						
Never tried	☐	☐	☐	☐	☐	Very easily
14. I can upgrade device software on a mobile device						
Never tried	☐	☐	☐	☐	☐	Very easily

A.4 Codebook for data analysis

Table 5: The codebook of relational maintenance strategies used in our data analysis, adapted from [9].

<i>Maintenance strategies</i>	<i>Definition</i>
Positivity	Attempts to make interactions cheerful and pleasant
Openness	Direct discussions and listening to one another
Assurances	Covertly and overtly assuring each other of their skills and abilities
Social Network	Relying on the support of friends and family
Shared Tasks	Performing routine tasks and chores in a relationship
Joint activities	Spending time with one another
Calls/Texts/Emails	Use of mediated communication
Avoidance	Evasion of interaction or issues
Anti-social	Behaviors which seem unfriendly or coercive
Humor	Jokes and sarcasm